

E020

Data and AI Strategy

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

At a regional insurer, the customer claims data lives in a cloud database run by the IT infrastructure team. The VP of Claims decides who may use the data and answers for its use. The database was migrated last year at a cost of \$400,000. Who is the data owner?

- A. A data steward in the claims department.
- B. The VP of Claims.
- C. The IT infrastructure team.
- D. The cloud provider, because it hosts the database.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which task belongs to a data custodian rather than a data owner?

- A. Deciding which departments may use the data.
- B. Setting the data's classification level.
- C. Defining what "active customer" means in the data.
- D. Running backups and applying the access controls the owner approved.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A hospital's patient portal shows allergy lists, but 30% of records have the allergy field blank even though the patients reported allergies at intake. The portal was redesigned with a new color scheme in March. Which data quality dimension is failing?

- A. Accuracy.
- B. Timeliness.
- C. Consistency, because the portal was redesigned.
- D. Completeness.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A retailer wants to store raw clickstream logs, product images, and call-center recordings so data scientists can explore them later for machine learning. The chief data officer has a background in marketing. Which storage approach fits?

- A. A data lake with schema-on-read.
- B. A spreadsheet library.
- C. A data warehouse with schema-on-write.
- D. A warehouse, because every analytics need requires cleaned data first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Finance needs a trusted monthly revenue report built from cleaned, structured sales data with fixed definitions. Which storage approach fits?

- A. A data lake with schema-on-read.
- B. A data warehouse with schema-on-write.
- C. An operational database, since it holds the same sales records.
- D. The raw transaction logs, queried directly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A grocery chain with 85 stores builds a model that recommends how much of each product to order per store next week to minimize waste and stockouts. The project was funded from the innovation budget. Which level of analytics is this?

- A. Prescriptive.
- B. Descriptive.
- C. Predictive.
- D. Diagnostic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A team wants to understand why customer returns doubled last quarter. Which type of analytics answers that question?

- A.** Descriptive.
- B.** Diagnostic.
- C.** Predictive.
- D.** Prescriptive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A fitness app plans a new feature to track workout streaks. The product lead proposes also collecting users' contacts and precise location all day "in case we find a use later." The app has a 4.7-star rating. What is the best data strategy response?

- A.** Collect it and anonymize it later.
- B.** Collect it, since users' high ratings show they trust the app.
- C.** Collect only what the streak feature needs, with clear notice and a lawful basis.
- D.** Collect it all but encrypt it, which removes privacy concerns.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A utility has two AI proposals. One uses five years of clean meter data to predict equipment failures that cost \$3 million a year. The other would build a chatbot on a new model the CEO read about, but customer-question data is scattered and unlabeled. The board meets on Thursdays. Which proposal should lead?

- A.** Both, split the budget evenly.
- B.** The chatbot, because the CEO is interested in the new model.
- C.** Neither, since AI should wait until all company data is perfect.
- D.** The failure-prediction model: clear value and data that is ready.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A bank tests a fraud model on 10,000 transactions, of which 100 are fraud. The model flags 80 transactions, and 60 of those are actual fraud. The vendor's demo ran on a GPU cluster in its Seattle office. What is the model's recall?

- A. 80%
- B. 75%
- C. 60%
- D. 99.4%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

11. [TRANSFER — reversed stem / different variable]

A screening model flags 50 records, of which 40 are true positives. There are 200 true positives in the data set. What is precision?

- A. 80%
- B. 20%
- C. 125%
- D. 25%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12.

A lender's AI model for loan pre-approval passed fairness tests at launch 18 months ago. Since then the applicant mix has shifted, and approval rates for one ZIP-code group have fallen sharply. The model was built in Python by a four-person team. What should governance require now?

- A. Remove ZIP code from reports so the gap is not visible.
- B. Rewrite the model in a different language.
- C. Re-test for bias and drift, with human review of affected decisions until resolved.
- D. Nothing; it passed fairness tests at launch.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

At a regional insurer, the customer claims data lives in a cloud database run by the IT infrastructure team. The VP of Claims decides who may use the data and answers for its use. ~~The database was migrated last year at a cost of \$400,000.~~ Who is the data owner?

Target word: identify the owner **Decides:** The business leader accountable for the data.

A	Adjacent category	Stewards manage quality and definitions, not accountability.
B	Correct	The accountable business leader who approves use is the owner.
C	Confusion set	They are the custodian: they run the storage.
D	Camouflage	Hosting location does not decide ownership.

2. Answer D Concept 1 • transfer

Which task belongs to a data custodian rather than a data owner?

Target word: reverse: pick the custodian duty **Decides:** Custodians carry out technical controls.

A	Inverted operation	That is the owner's decision.
B	Confusion set	Owners set classification.
C	Adjacent category	Business definitions are steward work.
D	Correct	Custodians operate the technical environment.

3. Answer D Concept 2

A hospital's patient portal shows allergy lists, but 30% of records have the allergy field blank even though the patients reported allergies at intake. ~~The portal was redesigned with a new color scheme in March.~~ Which data quality dimension is failing?

Target word: name the dimension **Decides:** Required values are missing.

A	Confusion set	Accuracy is wrong values; these are missing values.
B	Adjacent category	Timeliness is about data being current, not present.
C	Camouflage	The redesign does not decide the defect.
D	Correct	Blank fields that should hold values are a completeness failure.

4. Answer A Concept 3

A retailer wants to store raw clickstream logs, product images, and call-center recordings so data scientists can explore them later for machine learning. ~~The chief data officer has a background in marketing.~~ Which storage approach fits?

Target word: fit storage to data type and use **Decides:** Raw, varied data for later exploration.

A	Correct	Lakes hold raw, varied data and apply structure when read.
B	Adjacent category	Cannot hold or scale to images and recordings.
C	Confusion set	Warehouses need structured, modeled data up front.
D	Absolute language	Exploration and ML often start from raw data.

5. Answer B Concept 3 · transfer

Finance needs a trusted monthly revenue report built from cleaned, structured sales data with fixed definitions. Which storage approach fits?

Target word: reverse: curated reporting need **Decides:** Structured, modeled data for consistent reporting.

A	Inverted operation	Suited to raw exploration, not governed standard reports.
B	Correct	Warehouses enforce structure up front for reliable reporting.
C	Adjacent category	Built for transactions, not for historical analysis and reporting.
D	Dropped element	Skips the cleaning and fixed definitions the report needs.

6. Answer A Concept 4

A grocery chain ~~with 85 stores~~ builds a model that recommends how much of each product to order per store next week to minimize waste and stockouts. ~~The project was funded from the innovation budget.~~ Which level of analytics is this?

Target word: maturity level **Decides:** It recommends an action.

A	Correct	Recommending what to do is prescriptive.
B	Confusion set	Descriptive reports what already happened.
C	Adjacent category	Forecasting demand alone would be predictive; this goes on to recommend orders.
D	Inverted operation	Diagnostic explains past causes rather than guiding future action.

7. Answer B Concept 4 · transfer

A team wants to understand why customer returns doubled last quarter. Which type of analytics answers that question?

Target word: reverse: level from the question **Decides:** “Why” is diagnostic.

A	Dropped element	Shows returns doubled but not why.
B	Correct	Diagnostic analytics explains causes.
C	Inverted operation	Looks forward, not back at causes.
D	Adjacent category	Recommends actions; the question asks for causes.

8. Answer C Concept 5

A fitness app plans a new feature to track workout streaks. The product lead proposes also collecting users' contacts and precise location all day “in case we find a use later.” ~~The app has a 4.7 star rating.~~ What is the best data strategy response?

Target word: collection principle **Decides:** Extra data has no defined purpose.

A	Inverted operation	Puts the safeguard after the over-collection.
B	Camouflage	Ratings do not grant consent or purpose.
C	Correct	Data minimization and purpose limitation.
D	Absolute language	Encryption protects data; it does not justify collecting it.

9. Answer D Concept 6

A utility has two AI proposals. One uses five years of clean meter data to predict equipment failures that cost \$3 million a year. The other would build a chatbot on a new model the CEO read about, but customer-question data is scattered and unlabeled. ~~The board meets on Thursdays.~~ Which proposal should lead?

Target word: prioritize use case **Decides:** High value plus ready data vs. low data readiness.

A	Dropped element	Ignores the feasibility gap in the chatbot data.
B	Inverted operation	Starts from the technology, not the problem.
C	Absolute language	Perfect data is never required; readiness for the use case is.
D	Correct	Value × feasibility favors it.

10. Answer C Concept 7

A bank tests a fraud model on 10,000 transactions, of which 100 are fraud. The model flags 80 transactions, and 60 of those are actual fraud. ~~The vendor's demo ran on a GPU cluster in its Seattle office.~~ What is the model's recall?

Target word: recall **Decides:** TP 60 of 100 actual fraud cases.

A	Wrong stem number	Divides the flagged count by the fraud count.
B	Confusion set	That is precision: 60 / 80.
C	Correct	60 / 100 = 60%.
D	Adjacent category	That is accuracy: (60 + 9,880) / 10,000, inflated by the many non-fraud cases.

11. Answer A Concept 7 • transfer

A screening model flags 50 records, of which 40 are true positives. There are 200 true positives in the data set. What is precision?

Target word: solve for precision instead of recall **Decides:** Precision = TP / flagged.

A	Correct	40 / 50 = 80%.
B	Confusion set	That is recall: 40 / 200.
C	Inverted operation	Divides flagged by TP; precision cannot exceed 100%.
D	Wrong stem number	Divides flagged (50) by total positives (200).

12. Answer C Concept 8

A lender's AI model for loan pre-approval passed fairness tests at launch 18 months ago. Since then the applicant mix has shifted, and approval rates for one ZIP-code group have fallen sharply. ~~The model was built in Python by a four-person team.~~ What should governance require now?

Target word: ongoing governance action **Decides:** Performance and fairness can drift after launch.

A	Inverted operation	Hides the signal instead of addressing it.
B	Camouflage	The coding language does not decide fairness.
C	Correct	Monitoring and oversight must continue after deployment.
D	Absolute language	Launch results do not guarantee later behavior.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Data governance roles	The data owner is the business leader accountable for a data set; the data steward manages its quality and definitions day to day; the data custodian runs the technical storage and controls.	Making IT the owner of business data because IT stores it.	“Who is accountable / who approves access?”
2	Data quality dimensions	Accuracy, completeness, consistency, timeliness, validity, and uniqueness are separate dimensions; fix problems at the source, not only downstream.	Calling every problem “inaccurate” or cleaning the same errors every month in reports.	A described defect: missing fields, stale data, duplicates, conflicting values.
3	Warehouse vs. lake	A data warehouse stores structured, cleaned data with schema-on-write for reporting; a data lake stores raw data of any type with schema-on-read for exploration and ML.	Assuming a lake is just a bigger warehouse.	Raw, unstructured, or varied data vs. curated reporting.
4	Analytics maturity	Descriptive = what happened; diagnostic = why; predictive = what will happen; prescriptive = what to do.	Calling a forecast prescriptive, or a dashboard predictive.	The question the output answers.
5	Responsible data collection	Collect only what a stated purpose needs (minimization), with a lawful basis or consent, and use it only for that purpose.	Collecting everything “in case it is useful later.”	A plan to gather extra personal data without a defined use.
6	AI strategy alignment	Choose AI use cases by business value and feasibility (data readiness, skills, risk), starting from a problem, not a technology.	Starting with a trendy model and looking for a place to use it.	Competing use cases, or a proposal that names the tool before the problem.
7	Evaluating AI: precision, recall, accuracy	Precision = $TP / (TP + FP)$; recall = $TP / (TP + FN)$; accuracy can look high on imbalanced data even when the model misses most positives.	Mixing up precision and recall, or trusting accuracy on rare-event data.	Counts of flagged items, true cases, and a rare outcome.
8	AI governance and risk	High-impact AI decisions need bias testing, explainability suited to the use, human oversight, and ongoing monitoring for drift.	Assuming a model that tested well at launch stays fair and accurate.	An AI decision affecting people, or performance changing over time.

Course scope (public description): Data and AI Strategy will familiarize the student with aspects of data science and artificial intelligence within the scope of a technology leader’s role. The student will be asked to analyze two technology strategies, although they will not be asked to develop their own strategies or set up any data or AI technologies. The course covers skills in data management needs, how data is used in an organization, data collection, AI organizational strategies and alignment, how organizations evaluate AI, and how AI impacts

technologies within business operations. There are no prerequisites for this course.